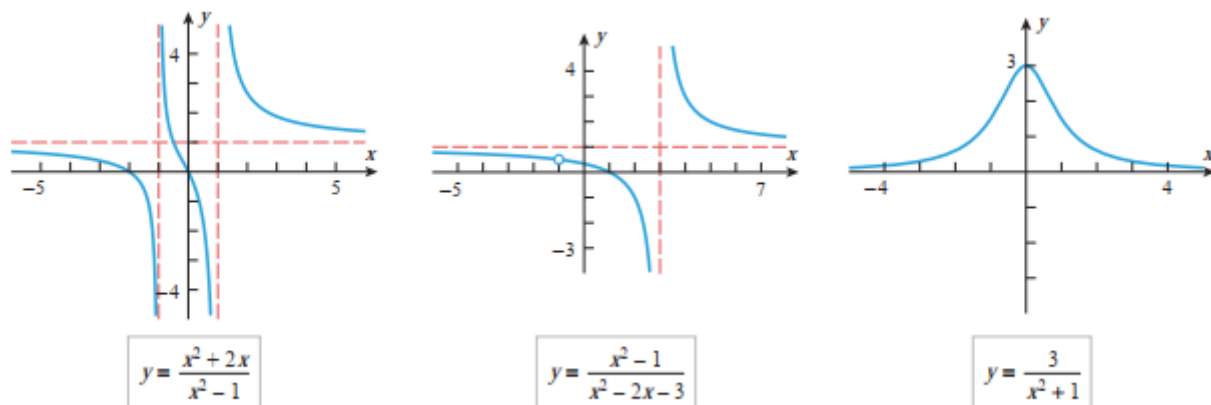


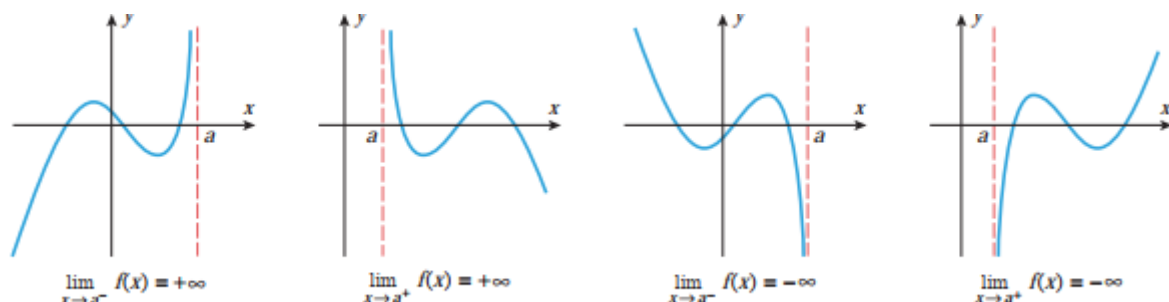
## Asymptotes of a function

- Unlike polynomials, rational functions may have numbers at which they are not defined. Near such points, many rational functions have graphs that closely approximate a vertical line, called a **vertical asymptote**. These are represented by the dashed vertical lines in Figure 0.3.11.
- Unlike the graphs of nonconstant polynomials, which eventually rise or fall indefinitely, the graphs of many rational functions eventually get closer and closer to some horizontal line, called a **horizontal asymptote**, as one traverses the curve in either the positive or negative direction. The horizontal asymptotes are represented by the dashed horizontal lines in the first two parts of Figure 0.3.11. In the third part of the figure the  $x$ -axis is a horizontal asymptote.



▲ Figure 0.3.11

## Vertical Asymptotes



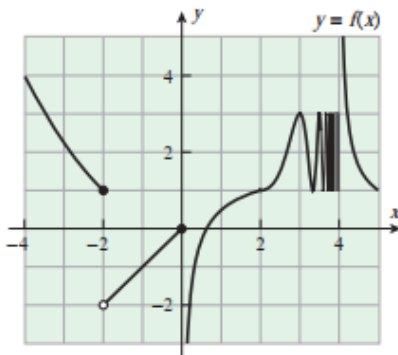
▲ Figure 1.1.17

Figure 1.1.17 illustrates geometrically what happens when any of the following situations occur:

$$\lim_{x \rightarrow a^-} f(x) = +\infty, \quad \lim_{x \rightarrow a^+} f(x) = +\infty, \quad \lim_{x \rightarrow a^-} f(x) = -\infty, \quad \lim_{x \rightarrow a^+} f(x) = -\infty$$

In each case the graph of  $y = f(x)$  either rises or falls without bound, squeezing closer and closer to the vertical line  $x = a$  as  $x$  approaches  $a$  from the side indicated in the limit. The line  $x = a$  is called a **vertical asymptote** of the curve  $y = f(x)$  (from the Greek word *asymptotos*, meaning “nonintersecting”).

In general, the graph of a single function can display a wide variety of limits.



▲ Figure 1.1.18

► **Example 8** For the function  $f$  graphed in Figure 1.1.18, find

- (a)  $\lim_{x \rightarrow -2^-} f(x)$  (b)  $\lim_{x \rightarrow -2^+} f(x)$  (c)  $\lim_{x \rightarrow 0^-} f(x)$  (d)  $\lim_{x \rightarrow 0^+} f(x)$   
 (e)  $\lim_{x \rightarrow 4^-} f(x)$  (f)  $\lim_{x \rightarrow 4^+} f(x)$  (g) the vertical asymptotes of the graph of  $f$ .

**Solution (a) and (b).**

$$\lim_{x \rightarrow -2^-} f(x) = 1 = f(-2) \quad \text{and} \quad \lim_{x \rightarrow -2^+} f(x) = -2$$

**Solution (c) and (d).**

$$\lim_{x \rightarrow 0^-} f(x) = 0 = f(0) \quad \text{and} \quad \lim_{x \rightarrow 0^+} f(x) = -\infty$$

**Solution (e) and (f).**

$$\lim_{x \rightarrow 4^-} f(x) \text{ does not exist due to oscillation} \quad \text{and} \quad \lim_{x \rightarrow 4^+} f(x) = +\infty$$

**Solution (g).** The  $y$ -axis and the line  $x = 4$  are vertical asymptotes for the graph of  $f$ . ◀

**Q.** Determine whether the following function has a vertical asymptote?

$$f(x) = \frac{1}{x-2}$$

**Solution:**

Since there is a discontinuity at  $x = 2$

∴ Taking  $\lim_{x \rightarrow 2}$  on both sides

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \frac{1}{x-2}$$

Applying limit

$$\lim_{x \rightarrow 2} f(x) = \frac{1}{2-2} = \frac{1}{0} = \infty$$

∴ there is a vertical asymptote at  $x = 2$

**Q.** Determine whether the following function has a vertical asymptote?

$$f(x) = \frac{x^2 - 1}{x - 1}$$

**Solution:**

Since  $f(x)$  is undefined at  $x = 1$

∴ Taking  $\lim_{x \rightarrow 1}$  on both sides

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$$

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{x-1}$$

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} (x + 1)$$

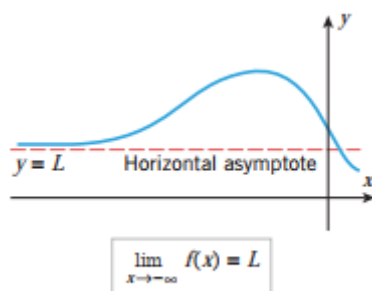
Applying limit

$$\lim_{x \rightarrow 1} f(x) = 1 + 1 = 2$$

Since  $\lim_{x \rightarrow 1} f(x) \neq \infty$

∴ there is no vertical asymptote in  $f(x)$

## Horizontal Asymptotes

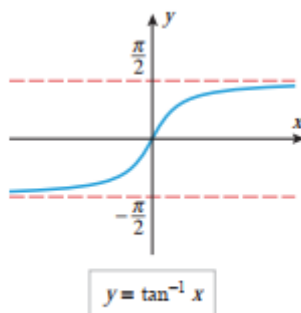


▲ Figure 1.3.2

Figure 1.3.2 illustrates the end behavior of a function  $f$  when

$$\lim_{x \rightarrow +\infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = L$$

In the first case the graph of  $f$  eventually comes as close as we like to the line  $y = L$  as  $x$  increases without bound, and in the second case it eventually comes as close as we like to the line  $y = L$  as  $x$  decreases without bound. If either limit holds, we call the line  $y = L$  a **horizontal asymptote** for the graph of  $f$ .

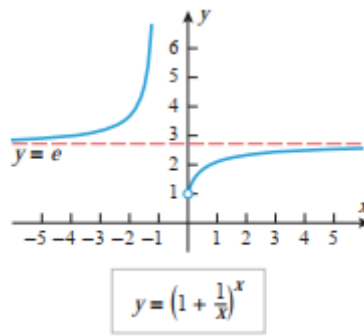


▲ Figure 1.3.3

► **Example 2** Figure 1.3.3 is the graph of  $f(x) = \tan^{-1} x$ . As suggested by this graph,

$$\lim_{x \rightarrow +\infty} \tan^{-1} x = \frac{\pi}{2} \quad \text{and} \quad \lim_{x \rightarrow -\infty} \tan^{-1} x = -\frac{\pi}{2} \quad (5-6)$$

so the line  $y = \pi/2$  is a horizontal asymptote for  $f$  in the positive direction and the line  $y = -\pi/2$  is a horizontal asymptote in the negative direction. ◀



▲ Figure 1.3.4

► **Example 3** Figure 1.3.4 is the graph of  $f(x) = (1 + 1/x)^x$ . As suggested by this graph,

$$\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x}\right)^x = e \quad \text{and} \quad \lim_{x \rightarrow -\infty} \left(1 + \frac{1}{x}\right)^x = e \quad (7-8)$$

so the line  $y = e$  is a horizontal asymptote for  $f$  in both the positive and negative directions. ◀

**Q.** Determine whether the following function has a horizontal asymptote?

$$f(x) = \frac{3x + 2}{x + 5}$$

Solution:

Taking  $\lim_{x \rightarrow \infty}$  on both sides

$$\begin{aligned} \lim_{x \rightarrow \infty} f(x) &= \lim_{x \rightarrow \infty} \frac{3x+2}{x+5} \\ \lim_{x \rightarrow \infty} f(x) &= \lim_{x \rightarrow \infty} \frac{x(3+\frac{2}{x})}{x(1+\frac{5}{x})} = \lim_{x \rightarrow \infty} \frac{(3+\frac{2}{x})}{(1+\frac{5}{x})} \end{aligned}$$

Applying limit

$$\lim_{x \rightarrow \infty} f(x) = \frac{3+\frac{2}{\infty}}{1+\frac{5}{\infty}} = \frac{3+0}{1+0} = 3$$

Taking  $\lim_{x \rightarrow -\infty}$  on both sides of  $f(x)$

$$\begin{aligned} \lim_{x \rightarrow -\infty} f(x) &= \lim_{x \rightarrow -\infty} \frac{3x+2}{x+5} \\ \lim_{x \rightarrow -\infty} f(x) &= \lim_{x \rightarrow -\infty} \frac{x(3+\frac{2}{x})}{x(1+\frac{5}{x})} = \lim_{x \rightarrow -\infty} \frac{(3+\frac{2}{x})}{(1+\frac{5}{x})} \end{aligned}$$

Applying limit

$$\lim_{x \rightarrow -\infty} f(x) = \frac{3+\frac{2}{-\infty}}{1+\frac{5}{-\infty}} = \frac{3-0}{1-0} = 3$$

Since  $\lim_{x \rightarrow \pm\infty} f(x) = 3$

∴ there is a horizontal asymptote at  $y = 3$

### Example 2 (two-sided behavior)

$$f(x) = \frac{x}{\sqrt{x^2 + 1}}$$

$$\lim_{x \rightarrow \infty} f(x) = 1, \quad \lim_{x \rightarrow -\infty} f(x) = -1$$

→ Asymptotes:

$$y = 1, \quad \downarrow \quad y = -1$$

### Practice Questions

Q1

$$f(x) = \frac{4x - 1}{2x + 3}$$

✓ Horizontal asymptote:  $y = 2$

Q2

$$f(x) = \frac{1}{x^2 - 4}$$

✓ Vertical asymptotes:  $x = \pm 2$

1. Find all asymptotes of:

$$f(x) = \frac{2x^2 + 3}{x - 1}$$

2. Using limits, show that  $y = 0$  is an asymptote of:

$$f(x) = xe^{-x}$$

3. Determine whether  $f(x) = \ln(x)/x$  has any asymptotes.